

Assignment 02

⇒ **Write a course report including the following issues.**

❖ Probability Basics

- Introduction to probability
- Additive and multiplicative rules with complements
- Types of events (independent, dependent, and mutually exclusive)
- Conditional probability
- Applications of Bayes' theorem

❖ Random Variables

- Introduction to random variables (definition and conditions)
- Types of random variables with examples
- Probability distributions functions
- Expectation and variance of the random variables
- Linear combinations for the expectation and variance of random variables
- Find the mean, median, quartiles, and mode

❖ Properties of Probability Distributions

- Discrete distributions: Binomial, Poisson
- Continuous Distributions: Normal

❖ Interval Estimation

- Confidence intervals for Mean
- Confidence intervals for Proportion

❖ Hypothesis Testing

- Introduction to hypothesis test (null and alternative hypotheses)
- One-tailed and two-tailed tests
- Type I and Type II Errors
- p –value